

Answer: D

$$24 \quad d \sin \theta = n \lambda$$

$$\left(\frac{1}{4.0 \times 10^5} \right) \sin \theta = 3 \times \left(\frac{3.00 \times 10^8}{6.0 \times 10^{14}} \right)$$

$$\theta = 37^\circ, 2\theta = 75^\circ$$

Answer: D

$$25 \quad \theta_1 = \theta_2 = \theta_3 = \frac{1}{2} \theta_4$$